

Network Vulnerability Scanning

Objective:

This task is performed to find vulnerabilities in a virtual network. I used Nmap for scan the local network. The scanning was performed for find open ports, detect OS, available vulnerabilities, analyse risk level.

Tools Used:

Nmap 7.95

Operating System: Kali Linux

Target Environment: VMware Lab Network

Targets Information:

Target IP Address	OS Detection
192.168.136.130	Linux Kernel 2.6.9 – 2.6.33
192.168.136.129	Microsoft Windows XP SP2

Scan Methodology:

These scans were performed to:

- Identify live hosts
- Discover open TCP ports
- Detect running services
- Identify the operating system

Open Ports and Services Identified:

1. Windows XP- This machine doesn't run lot of services. But some ports are opened.

```
(senthamilselvan@Kali)-[~]
$ nmap -sV 192.168.136.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-31 16:37 IST
Nmap scan report for 192.168.136.129
Host is up (0.00064s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows Vista Embedded microsoft-ds (workgroup: WORKGROUP)
1025/tcp   open  msrpc        Microsoft Windows RPC
3389/tcp   open  ms-wbt-server Microsoft Windows Terminal Service
MAC Address: 00:0C:29:9B:18:F2 (VMware)
Service Info: Host: JET-81F624C483C; OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_vista

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.69 seconds

(senthamilselvan@Kali)-[~]
```

2. Metasploitable- This machine runs lot of services and it has lot of open ports. The lot of ports are opened unwantedly.

```
Session Actions Edit View Help
(senthamilselvan@Kali)-[~]
$ nmap -sV 192.168.136.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-31 16:36 IST
Nmap scan report for 192.168.136.130
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp    open  rpcbind      2 (RPC #100000)
139/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp    open  exec         netkit-rsh rshexec
513/tcp    open  login        OpenBSD or Solaris rlogind
514/tcp    open  tcpwrapped
1099/tcp   open  java-rmi     GNU Classpath grmiregistry
1524/tcp   open  bindshell    Metasploitable root shell
2049/tcp   open  nfs          2-4 (RPC #100003)
2121/tcp   open  ftp          ProFTPD 1.3.1
3306/tcp   open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp   open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp   open  vnc          VNC (protocol 3.3)
6000/tcp   open  X11          (access denied)
6667/tcp   open  irc          UnrealIRCd
8080/tcp   open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp   open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:E6:0C:23 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.85 seconds

(senthamilselvan@Kali)-[~]
$
```

OS Detection Result:

- 192.168.136.129:
OS – Microsoft Windows 2003 | XP
Device Type – General Purpose
MAC Address – 00:0C:29:9B:18:F2

```

senthamilselvan@Kali: ~
Session Actions Edit View Help

(senthamilselvan@Kali)-[~]
$ nmap -O 192.168.136.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-31 16:29 IST
Nmap scan report for 192.168.136.129
Host is up (0.00066s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
1025/tcp   open  NFS-or-IIS
3389/tcp   open  ms-wbt-server
MAC Address: 00:0C:29:9B:18:F2 (VMware)
Device type: general purpose
Running: Microsoft Windows 2003|XP
OS CPE: cpe:/o:microsoft:windows_server_2003 cpe:/o:microsoft:windows_xp::sp2
OS details: Microsoft Windows Server 2003, Microsoft Windows Server 2003 SP1 or SP2, Microsoft Windows XP SP2 or Windows Server 2003 SP1 or SP2
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.49 seconds

(senthamilselvan@Kali)-[~]
$
```

- 192.168.136.130:
OS: Linux 2.6.X
Device Type: General Purpose
MAC Address: 00:0C:29:E6:0C:23

```

senthamilselvan@Kali: ~
Session Actions Edit View Help

(senthamilselvan@Kali)-[~]
$ nmap -O 192.168.136.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-31 16:30 IST
Nmap scan report for 192.168.136.130
Host is up (0.00064s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1009/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:E6:0C:23 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.76 seconds

(senthamilselvan@Kali)-[~]
$
```

Vulnerability Analysis:

The scan reveals several critical security issues. They are,
Telnet (Port 23) -transmits credentials in plaintext, making it vulnerable to interception.
FTP (Port 21) - may allow anonymous or weak authentication.

r-services (512–514) - rely on trust-based authentication and are highly insecure.

Ingreslock (1524) - is commonly associated with backdoor access.

SMB (445) - is a frequent target for remote exploitation and lateral movement.

Multiple database services (MySQL, PostgreSQL) are exposed without access restrictions.

Remote access services such as VNC and X11 are open, increasing attack surface.

Risk Assessment:

1.192.168.136.129 (XP):

Risk Level: Low

There is not many ports are opened. But the Operating System version is old one. So, it has changes to vulnerable.

2.192.168.136.130 (Linux):

Risk Level: Critical

There is lot of outdated services and many ports are opened. It makes the system vulnerable.

Recommendations:

To improve security posture, the following actions are recommended:

- Disable Telnet and replace it with SSH
- Disable unused services such as r-services and IRC
- Restrict database access to trusted IP addresses only
- Apply OS and kernel security patches
- Implement firewall rules to block unnecessary ports
- Use strong authentication mechanisms
- Regularly perform vulnerability scans